

Opal glass method, an extraction-free way to access microalgae pigment content

Supplementary materials

Victor Pozzobon , Clarisse Arnoudts, and Wendie Levasseur

LGPM, CentraleSupélec, Université Paris-Saclay, Centre Européen de Biotechnologie et de Bioéconomie (CEBB), 3 rue des Rouges Terres 51110 Pomacle, France
 victor.pozzobon@centralesupelec.fr

1. Effect of the cuvette type

As the method involves light scattering, the overall system geometry can be expected to impact the obtained spectra. Therefore, a test was led to assess the effect of the type of cuvette (3 mL and 1 mL, 10 mm light path in any case). The idea was two-fold: determine the system offering the best signal, and advise on whether the cuvette types are interchangeable. Figure 1 displays the spectra for a given sample measured with one cuvette type or the other, and baseline acquisition performed with one cuvette type or the other. As one can see, all the possible combinations yield different spectra. Therefore, cuvette types are not interchangeable. Then comes the question of the best setup. In this case, the highest signal-to-noise ratio was obtained for 3-mL cuvettes. They were thus retained for the rest of the experiments.

2. Spectra acquisition resolution

In order to find the best compromise between spectra acquisition time (the lower, the better) and data quality, all the spectral resolutions available on the spectrophotometer at hand (Shimadzu UV-1800 UV/Vis, dual beam) were tested. As one can see on Figure 2, modulating the resolution from 0.1

to 2 nm does not alter the quality of the obtained spectra, while reducing the acquisition time from 3 min 20 to 20 seconds. Therefore, a 2 nm resolution was opted for. Also, no matter the chosen resolution, the spectrum is plagued by a high level of noise in the 340-400 nm region. While it was still recorded, it will be omitted for legibility reasons in all the other graphs.

3. Method linearity

As for any spectrophotometric method, the linearity of the signal is to be checked. The primary objective is to ensure that high-quality data are generated. The secondary objective is to have a validity criterion that is easy to formulate and use on the field. Figure 3 presents the spectra obtained for a given sample applying various dilution rates. Here, the absolute dilution factor values are of limited interest *per se*. On the contrary, the focus is set on the agreement between the measurements. As one can see, the scaled spectra obtained for dilutions of 3 and 5 are satisfactorily close. Therefore, it could be concluded that a maximum absorbance value of 1.0 in the 400-800 nm region is advisable. As a safeguard procedure, a flexible threshold of 0.5 ± 0.1 was retained instead,

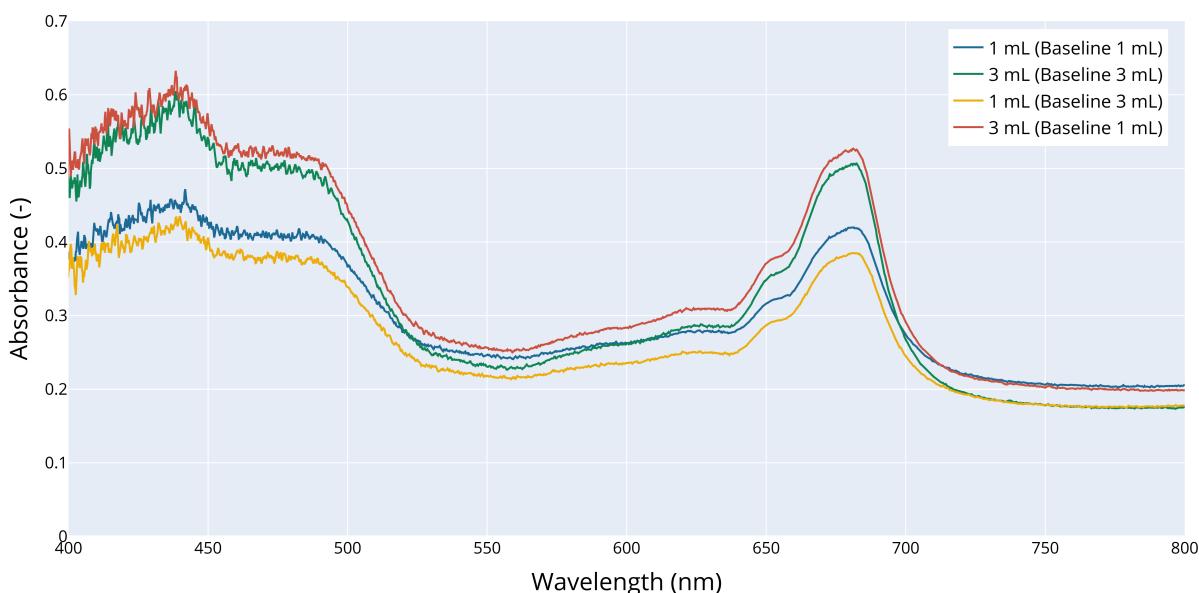


Fig. 1. Spectra obtained for a given sample measured in 3 and 1 mL cuvettes with baselines performed with 3 and 1 mL cuvettes

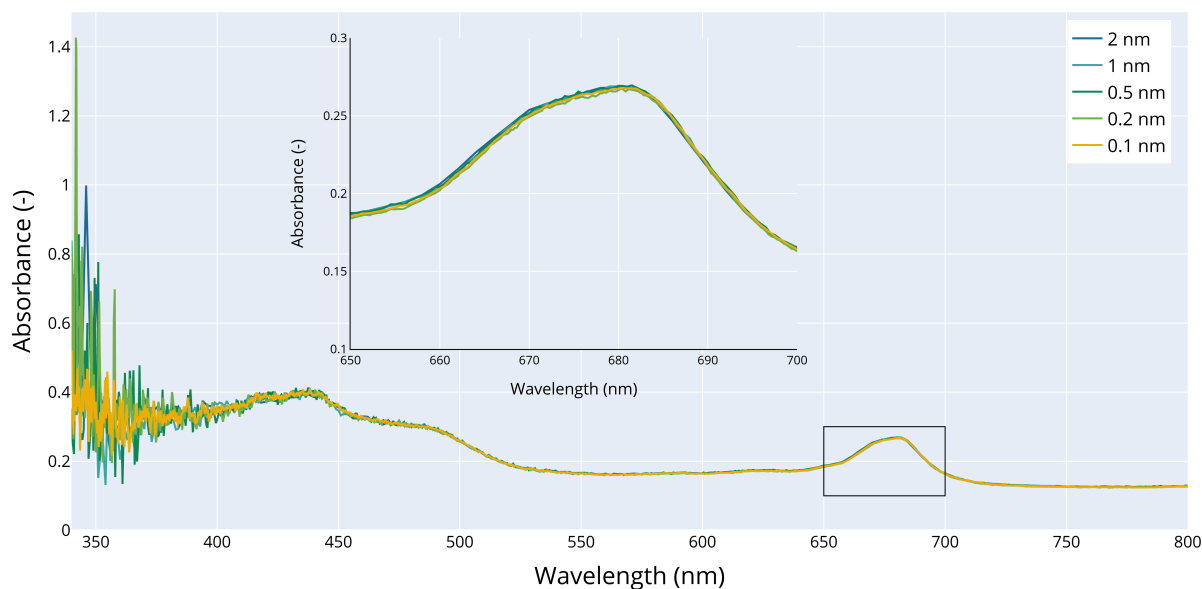


Fig. 2. Raw opal spectra obtained with different resolution for the same sample, analyzed sequentially and extemporaneously. Inset - focus on the high interest region that is the chlorophylls' peak

to be on the safe side.

4. Method repeatability

Until now, the same sample was used, but under different conditions (cuvette, dilution rate). It is also important to verify that for a given sample, consecutive measurements yield the same spectra. Figure 4 presents the results of this procedure applied to a sample for two different dilution factors. As one can see, the signal is very repeatable in the 500-800 nm region. On the contrary, in the 400-500 nm the noise appears to be random, hence different at each measurement appears a bit different. Furthermore, this noise scales with the signal value and may therefore originate from the sample state and not the scattering device. In any case, the process can be deemed repeatable.

5. Spectra filtration and derivation

While the level of noise generated by the method can be deemed acceptable in the red region of the chlorophyll, it may pose a problem on the blue end of the visible spectrum. It is all the more true as derivation is a must-include preprocessing technique for anyone willing to use spectral information. Therefore, a filtration preprocessing step was included in the workflow. The retained method was the Savitzky-Golay algorithm owing to its flexibility and widespread use. Two parameters can be adjusted: the polynomial order and fitting window size. Therefore, in order to identify the best pair, polynomial orders from 1 to 3 were tested in combination with a fitting window size ranging from 5 to 15 points. Figure 5 illustrates the results of this screening. Regarding signal filtration, first-order polynomials systematically induce signal shifts and should be discarded. On the contrary, orders 2 and 3 deliver excellent results until a window size of 9 points. Turning to the first-order derivative, order-2 poly-

mial expressions achieve good results up to a window size of 7, maybe 9. Order 3 expressions require at least 11 points, 13 being better, while exhibiting a lower sensitivity to the window size compared to their order 2 counterparts. Moving to order 2 derivation, for both order 2 and 3, a 13-point or more fitting window is required.

Consequently, filtration was carried out with an order 2 polynomial expression and a window size of 7 points. First-order derivation was carried out with an order 3 polynomial expression and a window size of 13 points. Second-order derivation was carried out with an order 3 polynomial expression and a window size of 13 points.

6. Opal glass vs. BSA - Baseline correction

As mentioned in the manuscript body, each test involving BSA addition required its own baseline. While the results in the manuscript body are presented with the baseline correction, Figure 6 presents the raw measurements. For BSA only measurements, the spectra in the 400-500 nm region exhibit chaotic behavior not congruent with the rest of the spectral domain (spectra "crossing" one another). The spectra obtained in conjunction with the opal glass method did not feature such a behavior. Yet, they also exhibit a hard-to-explain divergent trend in the blue end of the spectral domain. Finally, as one can see in the manuscript body, correcting for the BSA-induced baseline solved these problems.

7. PLS components

The number of components is the main hyperparameter to adjust to ensure proper functioning of the algorithm. Consequently, a cross-validation (5-fold) was carried out over the training set to obtain the residual error (MAPE) average and standard deviation for all the possible preprocessing combinations. The results are presented in Figure 7. As

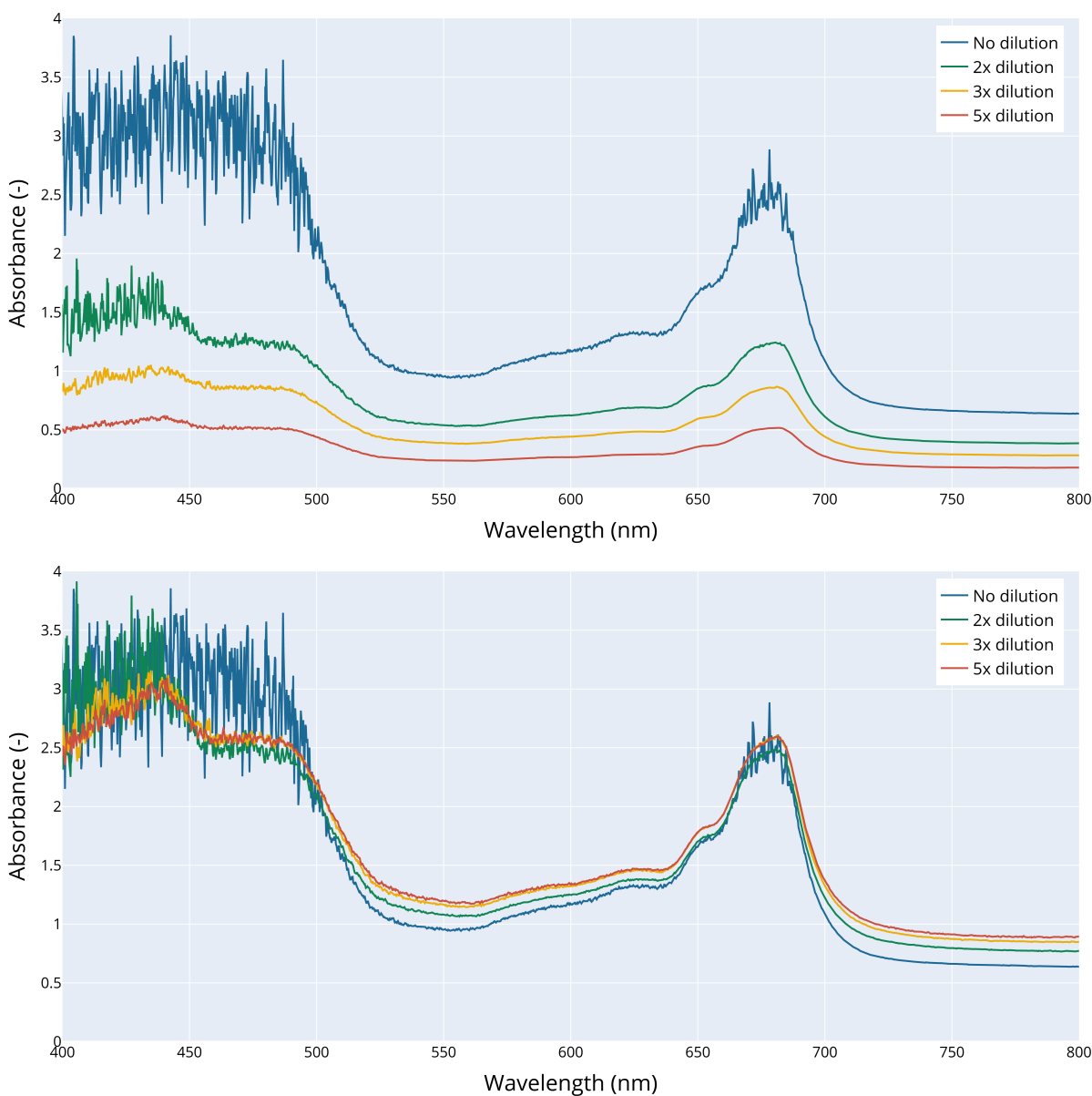


Fig. 3. Top - Raw signal obtained for a given sample at various dilution factors. Bottom - Scaled signal (raw signal $\times$ dilution factor) for a given sample at various dilution factors

one can see, the preprocessing methods impact the results. MSC and Z-scaling exhibit similar trends with high dispersion and marginal gains followed by hindered performance as the number of components increases. In contrast, not preprocessing the signal, or using the pigment-scaling procedure, yields curves with a characteristic elbow shape. The optimal number of components is 2 for all the outputs. Furthermore, pigment-scaling delivers marginally better results than the unprocessed signal (-2.44 ± 2.44 MAPE). Finally, filtering the signal offers benefit when a high number of components is used (configurations which are not relevant), but does not improve predictions when an adequate number of components is selected (-0.049 ± 0.093 MAPE).

8. CNN architectures

Figures 8 and 9 present the general architecture of the two neural networks used in this study. For each of the dimensions (three pigments and the total pigment content) and the preprocessings, the best set of parameters identified by Optuna (search-space in Table 1 and 3) for the internal layers is reported in Table 2 and 4. CNNs were trained using the Adam optimizer, with a decreasing learning rate (divided by two) when validation loss stalled for 50 epochs or more. In addition, to prevent overfitting, an early-stopping criterion was employed when the validation loss did not decrease significantly (0.01 MAPE) over 100 epochs.

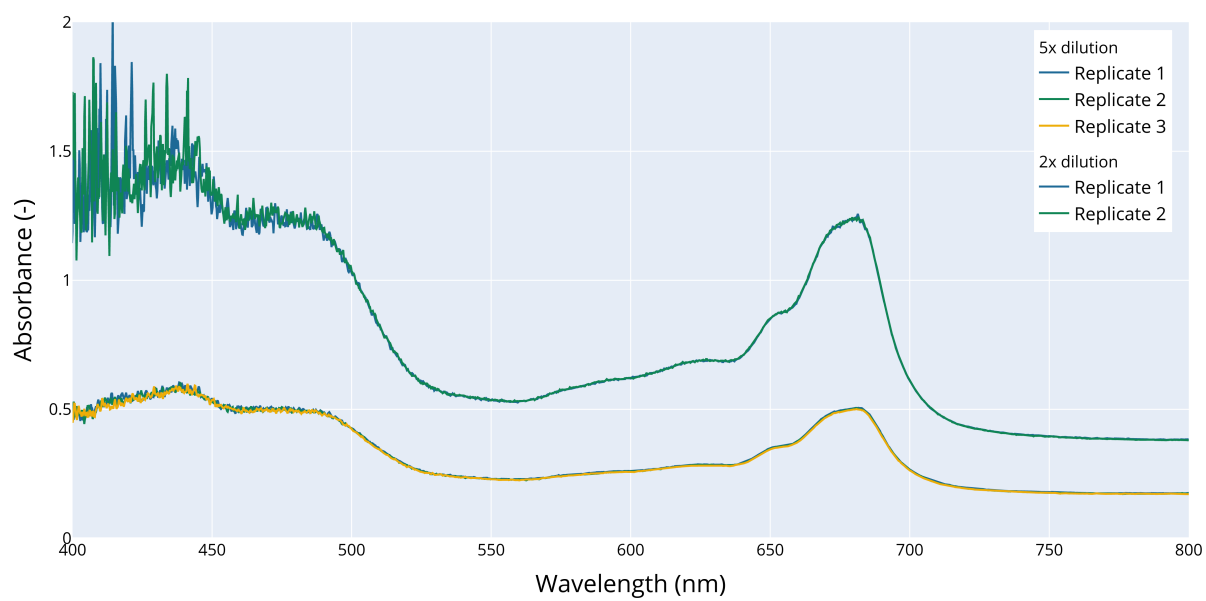


Fig. 4. Spectra obtained for consecutive measurements of a given sample for two dilution factors

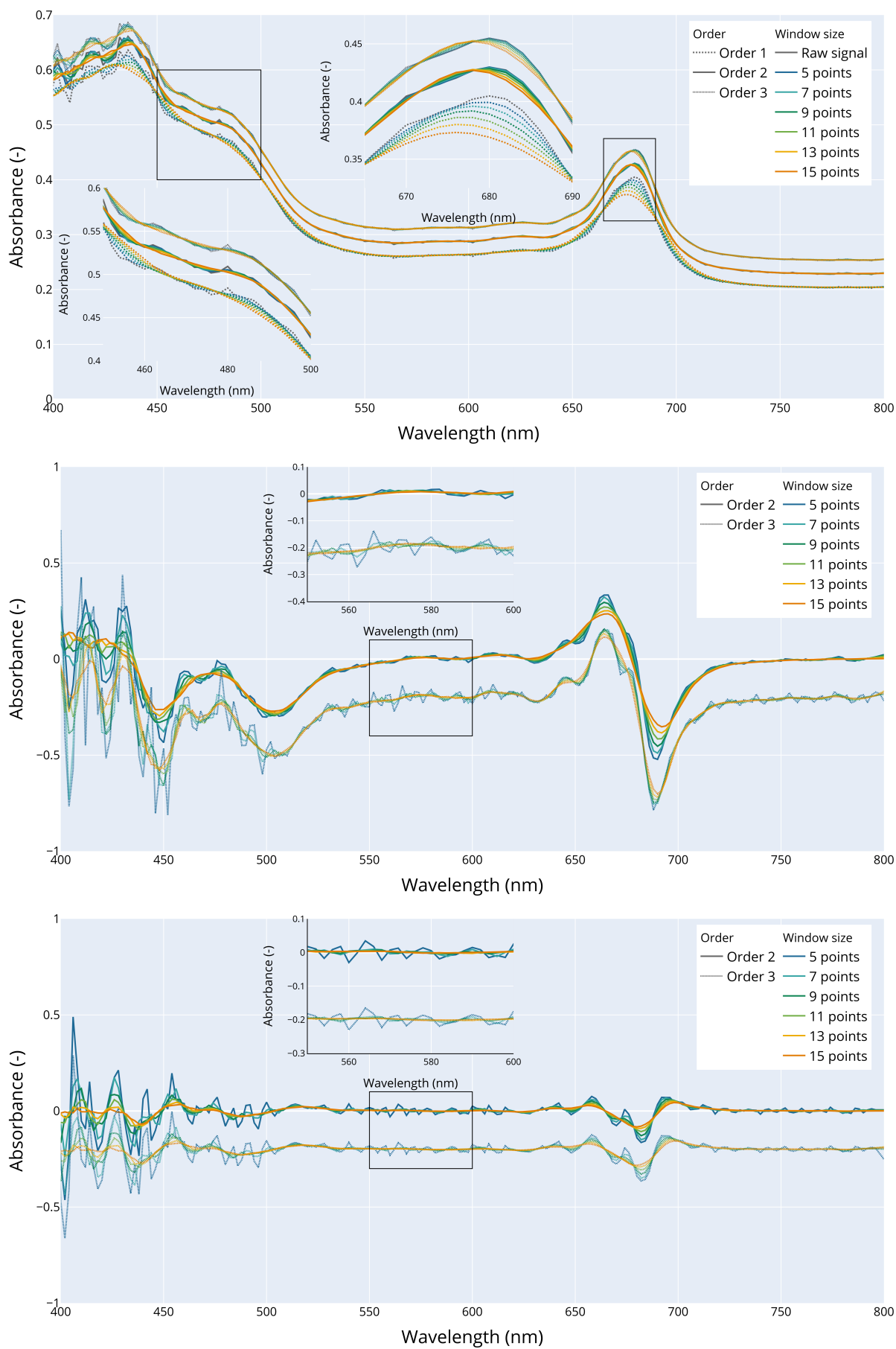


Fig. 5. Signals obtain after Savitzky-Golay processing. Top - Filtration. Middle - First-order derivative (signal multiplied by 25). Bottom - Second-order derivative (signal multiplied by 25). The vertical shift between the signal is added to improve legibility
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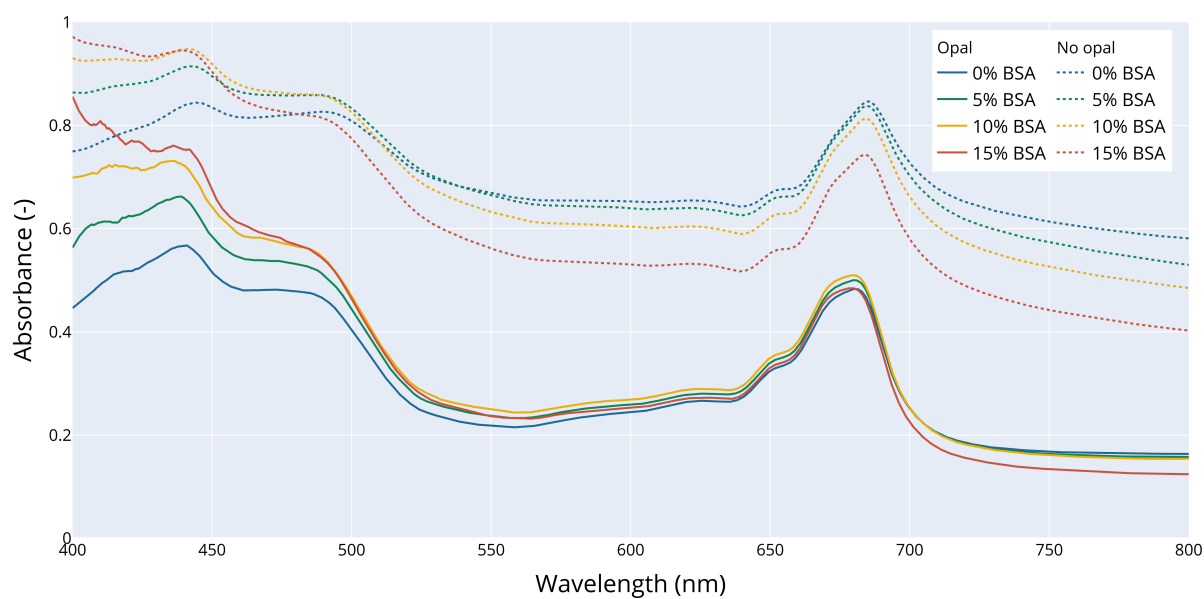


Fig. 6. Raw spectra obtained of the test comparing BSA loading and opal glass technique

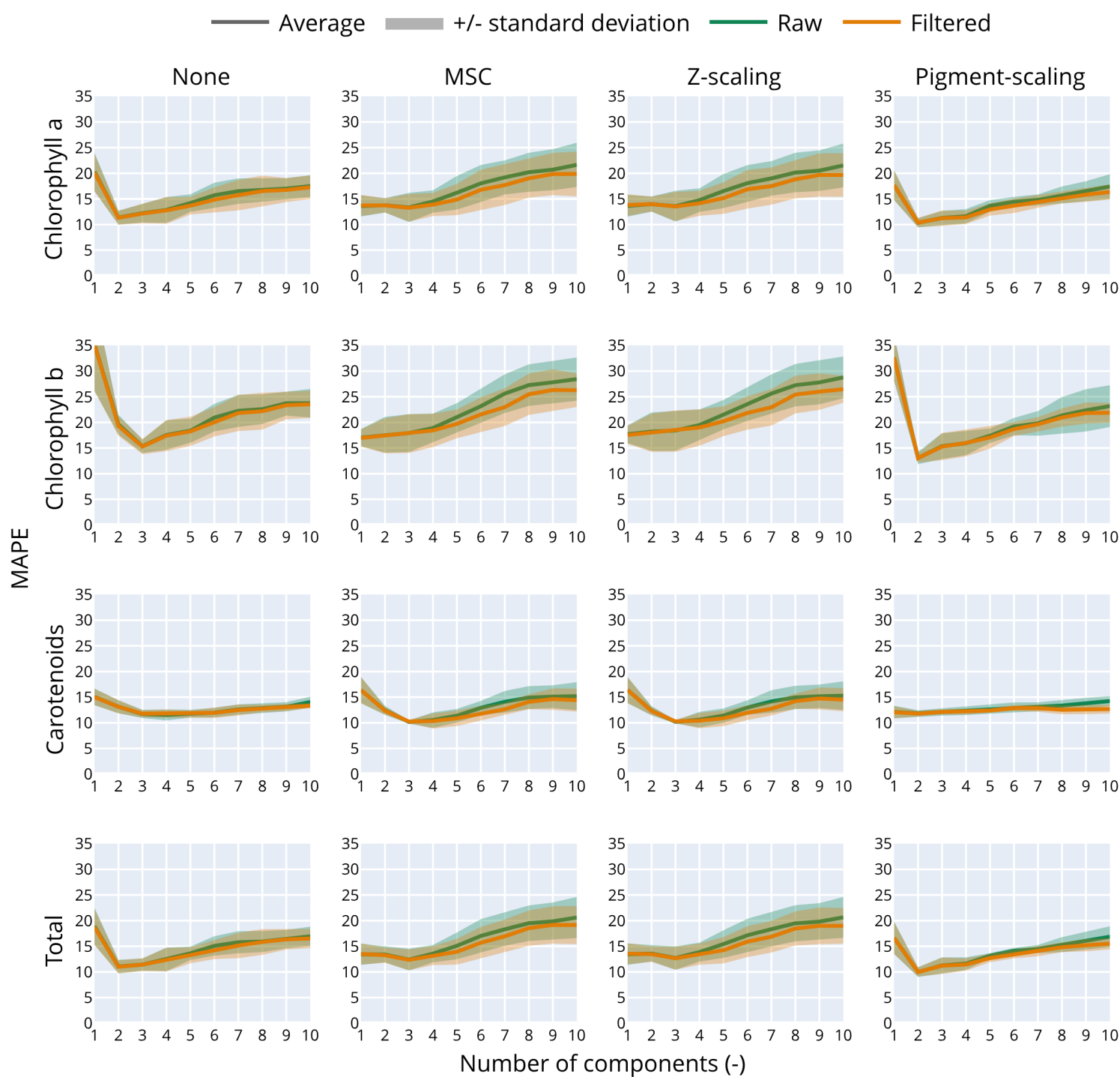


Fig. 7. Screen plot resulting from the screening of 1 to 10 components for all the possible preprocessing combinations. Solid line - Average of 5-fold cross-validation. Shading - Standard deviation obtained from 5-fold cross-validation

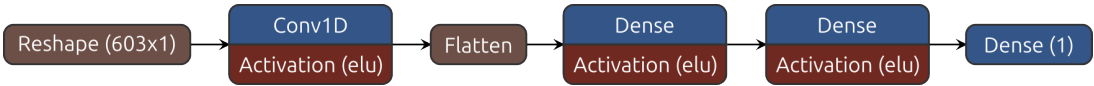


Fig. 8. Architectures of the Cui-style neural networks used in the study

Hyperparameter	Filter number	Kernel width	Kernel stride	Dense layer size (1)	Dense layer size (2)	Base learning rate	L2 regularization
Lower bound	1	1	1	1	1	$1 \cdot 10^{-4}$	$1 \cdot 10^{-5}$
Higher bound	Nb. features divided by 2	Nb. features divided by 5	Nb. features divided by 5	Nb. features	Nb. features divided by 2	0.1	0.1
Suggestion pattern	Linear	Linear	Linear	Linear	Linear	Linear in log. scale	Linear in log. scale

Table 1. Hyperparameters search range for Cui-style network

Preprocessing	Dimension	Filter number	Kernel width	Kernel stride	Dense layer size (1)	Dense layer size (2)	Base learning rate ($\times 1000$)	L2 regularization ($\times 1000$)	MAPE
None – Filtered	Chlorophyll a	43	36	56	558	172	2.312	4.805	<u>6.90</u>
	Chlorophyll b	31	58	48	459	218	14.942	1.531	<u>8.34</u>
	Carotenoids	37	34	50	342	204	0.887	21.642	<u>8.22</u>
	Total pigments	54	29	50	543	154	2.202	3.613	<u>7.18</u>
MSC – Filtered	Chlorophyll a	33	37	14	493	3	1.548	0.330	7.95
	Chlorophyll b	51	58	50	93	231	37.769	3.996	10.58
	Carotenoids	58	42	35	440	283	0.729	30.436	8.80
	Total pigments	38	44	20	107	34	67.996	0.016	8.30
Z-scaling – Filtered	Chlorophyll a	35	24	42	83	29	59.042	0.017	8.46
	Chlorophyll b	43	28	24	153	223	21.302	0.048	10.59
	Carotenoids	32	29	41	502	14	2.392	0.569	9.36
	Total pigments	40	57	20	538	257	39.145	0.485	8.27
Pigment-scaling – Filtered	Chlorophyll a	56	37	43	13	292	31.429	0.071	7.02
	Chlorophyll b	57	45	57	383	229	20.322	0.197	8.89
	Carotenoids	42	43	57	578	125	7.924	38.809	8.37
	Total pigments	59	42	58	21	276	30.294	0.026	7.32
None – Raw	Chlorophyll a	34	59	56	58	295	36.458	0.021	7.17
	Chlorophyll b	57	41	56	154	198	27.550	0.475	8.70
	Carotenoids	48	28	50	199	184	12.409	8.216	8.55
	Total pigments	30	38	60	416	161	9.642	0.016	7.24
MSC – Raw	Chlorophyll a	31	60	40	449	257	31.283	0.575	8.37
	Chlorophyll b	47	45	25	238	94	41.989	0.061	10.69
	Carotenoids	43	42	41	493	233	0.558	34.510	9.06
	Total pigments	37	56	52	46	31	91.324	0.748	8.44
Z-scaling – Raw	Chlorophyll a	53	55	44	309	104	56.106	0.498	8.22
	Chlorophyll b	58	44	21	370	190	26.458	0.165	10.72
	Carotenoids	51	30	47	441	278	0.253	0.627	9.76
	Total pigments	42	60	56	36	72	55.950	0.014	8.71
Pigment-scaling – Raw	Chlorophyll a	40	40	56	15	235	37.837	0.035	7.55
	Chlorophyll b	56	38	56	390	104	24.115	0.186	9.12
	Carotenoids	56	39	58	506	167	14.443	25.187	8.63
	Total pigments	48	43	55	74	47	45.430	0.132	7.43

Table 2. Specific network structure obtained after Optuna optimization for Cui-style network. Bold-underlined - Best value

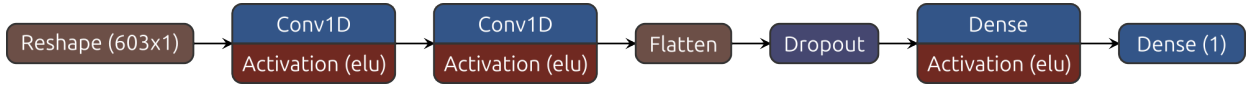


Fig. 9. Architectures of the Bjerrum-style neural networks used in the study

Hyperparameter	Filter number (1)	Kernel width (1)	Kernel stride (1)	Filter number (2)	Kernel width (2)	Kernel stride (2)	Dropout	Dense layer size	Base learning rate	L2 regularization
Lower bound	1	1	1	1	1	1	$1 \cdot 10^{-3}$	1	$1 \cdot 10^{-4}$	$1 \cdot 10^{-5}$
Higher bound	Nb. features divided by 2	Nb. features divided by 5	Nb. features divided by 5	Nb. features divided by 2	Nb. features divided by 5	Nb. features divided by 5	0.2	Nb. features	0.1	0.1
Suggestion pattern	Linear	Linear	Linear	Linear	Linear	Linear	Linear in log. scale	Linear	Linear in log. scale	Linear in log. scale

Table 3. Hyperparemeters search range for Bjerrum-style network

Preprocessing	Dimension	Filter number (1)	Kernel width (1)	Kernel stride (1)	Filter number (2)	Kernel width (2)	Kernel stride (2)	Dropout	Dense layer size	Base learning rate (x1000)	L2 regularization (x1000)	MAPE
None – Filtered	Chlorophyll a	51	29	33	33	16	14	2.89	96	1.010	8.225	7.09
	Chlorophyll b	57	6	32	57	13	50	0.29	74	6.317	5.957	<u>7.96</u>
	Carotenoids	57	53	26	49	35	1	6.49	524	4.523	23.745	<u>8.00</u>
	Total pigments	32	11	12	51	27	16	8.14	577	1.616	0.967	7.22
MSC – Filtered	Chlorophyll a	34	50	40	36	27	17	10.58	144	5.251	18.278	8.13
	Chlorophyll b	34	42	16	51	47	55	5.77	128	2.666	0.015	11.14
	Carotenoids	59	22	48	54	17	58	0.18	436	0.442	23.556	9.21
	Total pigments	51	57	40	38	52	52	9.95	25	7.198	0.191	8.28
Z-scaling – Filtered	Chlorophyll a	31	44	19	33	5	19	4.25	374	8.642	0.121	8.21
	Chlorophyll b	38	52	9	33	9	35	4.99	293	9.505	1.083	9.89
	Carotenoids	34	60	11	60	29	1	2.45	313	0.160	0.011	9.32
	Total pigments	30	9	32	42	48	22	1.54	471	1.982	0.446	8.31
Pigment-scaling – Filtered	Chlorophyll a	49	50	36	39	51	13	10.05	139	1.397	0.273	7.31
	Chlorophyll b	42	27	5	54	15	55	13.78	199	3.361	14.154	8.84
	Carotenoids	31	58	52	46	11	14	1.72	455	3.101	1.058	8.70
	Total pigments	40	30	11	56	15	41	5.28	23	3.073	0.019	7.26
None – Raw	Chlorophyll a	47	30	57	58	46	34	0.38	235	3.703	0.015	<u>7.05</u>
	Chlorophyll b	32	53	20	49	8	45	0.36	576	9.698	0.129	8.67
	Carotenoids	47	53	35	55	4	50	1.73	476	3.154	0.043	8.63
	Total pigments	30	1	9	38	43	18	0.13	511	7.539	1.347	<u>7.12</u>
MSC – Raw	Chlorophyll a	56	52	7	47	24	51	2.87	70	2.674	25.786	8.20
	Chlorophyll b	53	31	17	53	31	29	0.54	3	5.736	15.693	11.43
	Carotenoids	38	36	34	54	19	45	3.05	240	1.175	71.203	9.14
	Total pigments	57	41	13	38	8	24	1.17	441	8.234	4.031	8.48
Z-scaling – Raw	Chlorophyll a	55	34	18	39	7	20	2.64	99	6.249	59.580	8.35
	Chlorophyll b	48	15	1	53	33	10	0.52	305	5.181	1.681	10.41
	Carotenoids	38	20	23	60	26	21	0.74	341	0.290	27.454	9.62
	Total pigments	30	39	11	51	9	28	17.54	477	8.842	1.497	8.26
Pigment-scaling – Raw	Chlorophyll a	42	28	58	51	36	8	0.65	315	1.313	0.087	7.50
	Chlorophyll b	60	1	1	40	14	3	0.58	399	6.850	0.016	8.84
	Carotenoids	41	32	3	34	47	4	3.60	526	0.184	0.044	8.93
	Total pigments	59	25	20	31	9	41	0.25	97	9.009	0.180	7.52

Table 4. Specific network structure obtained after Optuna optimization for Bjerrum-style network. Bold-underlined - Best value

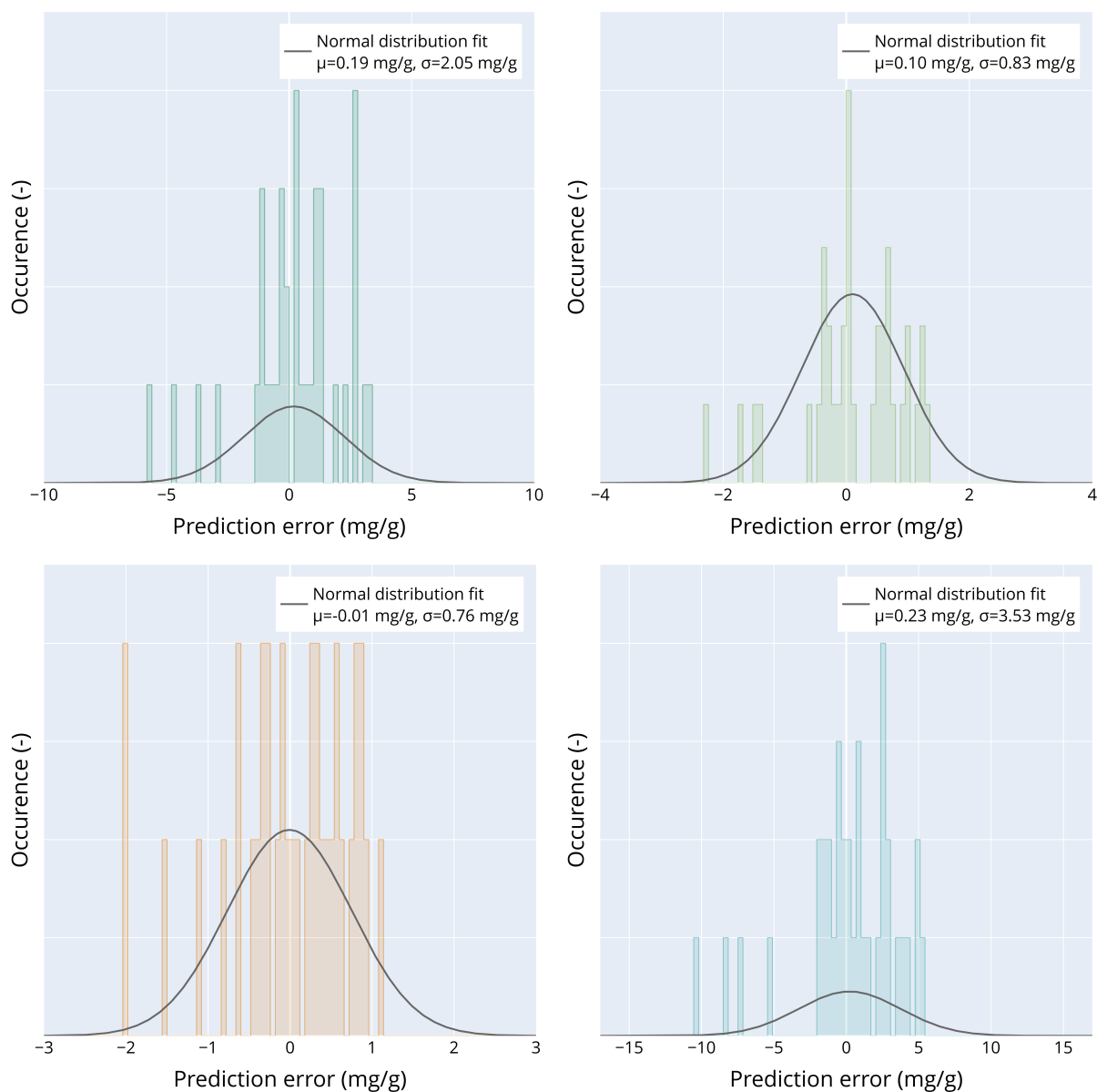


Fig. 10. Error plot on the validation set for the PLS (filtered and pigment-scaling) pipeline. Top left - Chlorophyll a. Top right - Chlorophyll b. Bottom left - Total carotenoids. Bottom right - total pigment

9. Error analysis

While parity plots and performance for the field trial are presented in the manuscript, they do not suffice to fully characterize the performance of the proposed pipelines. Indeed, it is also important to analyze the residuals/errors. Figure 10 and 11 present the histograms of the residuals for the PLS (filtered and pigment-scaling) and the CNN (Bjerrum architecture, filtered data, no additional preprocessing). In addition to the distribution itself, a fit using a normal distribution is proposed. As one can see, while the spread of the errors can vary, they can be considered relatively centered.

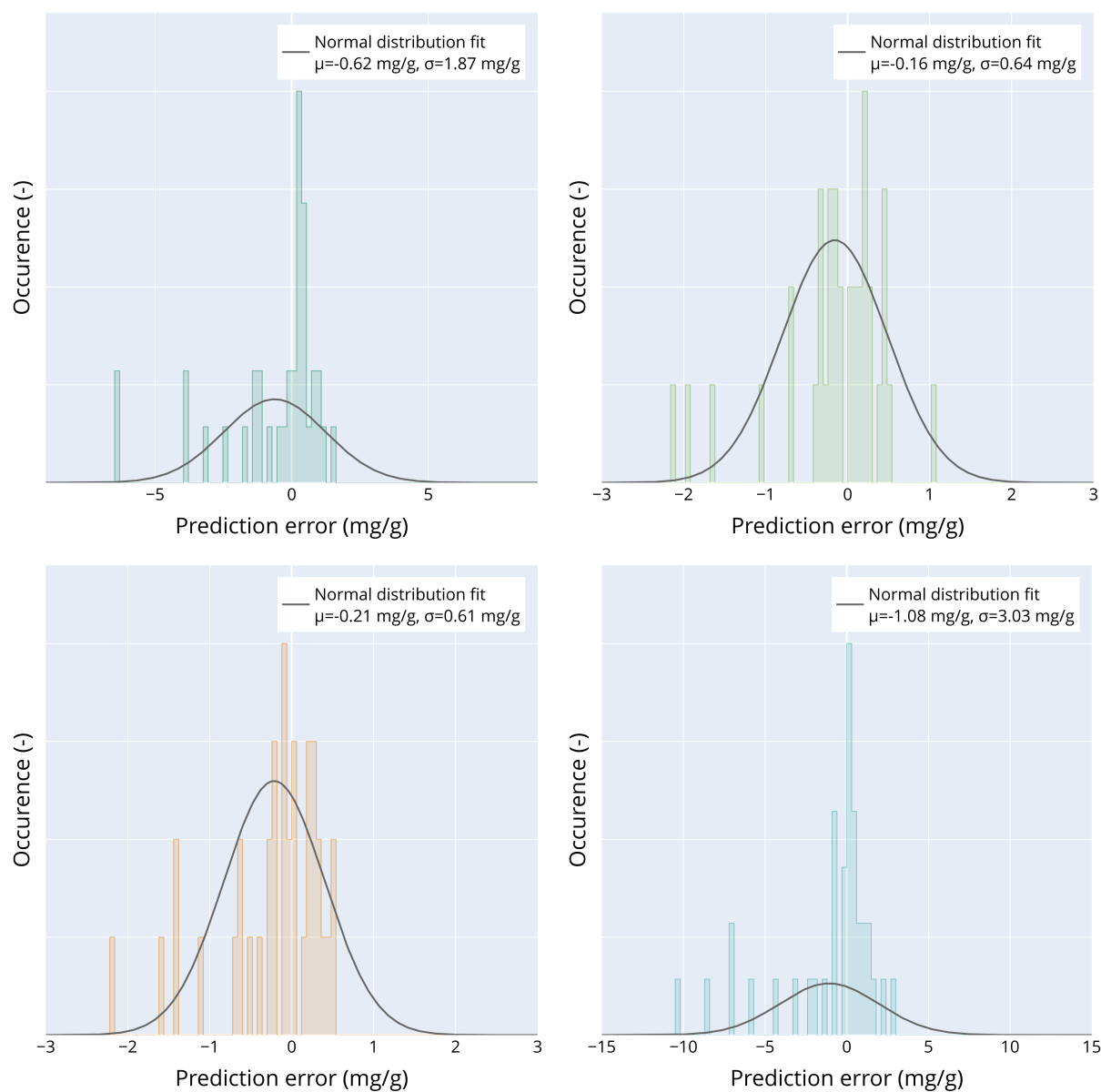


Fig. 11. Error plot on the validation set for the CNN (Bjerrum architecture, filtered data, no additional preprocessing) pipeline. Top left - Chlorophyll a. Top right - Chlorophyll b. Bottom left - Total carotenoids. Bottom right - total pigment